From Studio to Screen

A Roadmap for Indie Artists to Land Film & TV Placements

James Lai | Data Analytics Capstone | February 2026

Agenda

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The Revenue Gap

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The Revenue Gap

Streaming Revenue		Sync Licensing Revenue
+ ~166,667 – 1,666,667 Streams	GAP	+ ONE Sync Placement
$500 - $5,000		$500 - $5,000

“Sync is not just extra income, it is the foundation of a sustainable career.”

Step 1: The Problem

For 1,000-songs catalog,

Single-Emotion Precision

Supervisors search by **mood,** not **artist name**

~110 hours of labor

It is like a **full time job** for **3 weeks** straight.

The $300 Barrier

For Commercial AI tools (like Cyanite), that's **100,000 streams** to begin tagging.

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Step 2: The Question

Can a free, open-source Machine Learning replicate commercial emotion tagging at **>= 85% accuracy**?

The Goal

Giving artists the same discovery tools that big labels use.

Decode the “Vibe”

Translating raw audio into **actionable listener insights**

98% Time-Save

110 hours of manual labor reduced to **under 2 hours**.

Zero Gatekeeping

Pro-grade accuracy with **full transparency** and **$0 cost**.

The Data

278,000 Tracks

A massive open-source foundation for professional-grade statistical significance

8 Audio Metrics

Deep-dive analysis of **Tempo**, **Valence**, **Energy** and more.

4 Core Emotions

Mapping to **Sad**, **Happy**, **Energetic** and **Calm**

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Model Comparison

One rule. One hundred experts. See the difference.

	Logistic Regression	Random Forest Classifier
How it thinks	One straight line rule	100 experts voting
Complex Music	Get confused	Handles it
Tells you why	Not really	Always
Accuracy	**81% (below target)**	**91% (exceed target)**

Step 3: The Solution

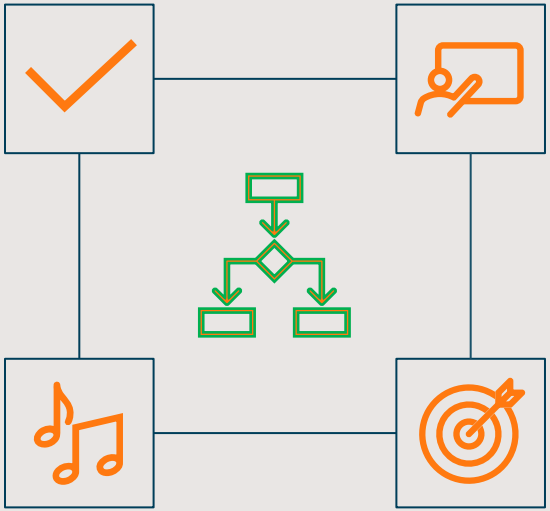
Random Forest Classifier: The Consensus Engine

100 “Expert Voters”

100 people are harder to fool than one.

Total Transparency

You will always know exactly why a song is tagged “Happy” or “Calm”.



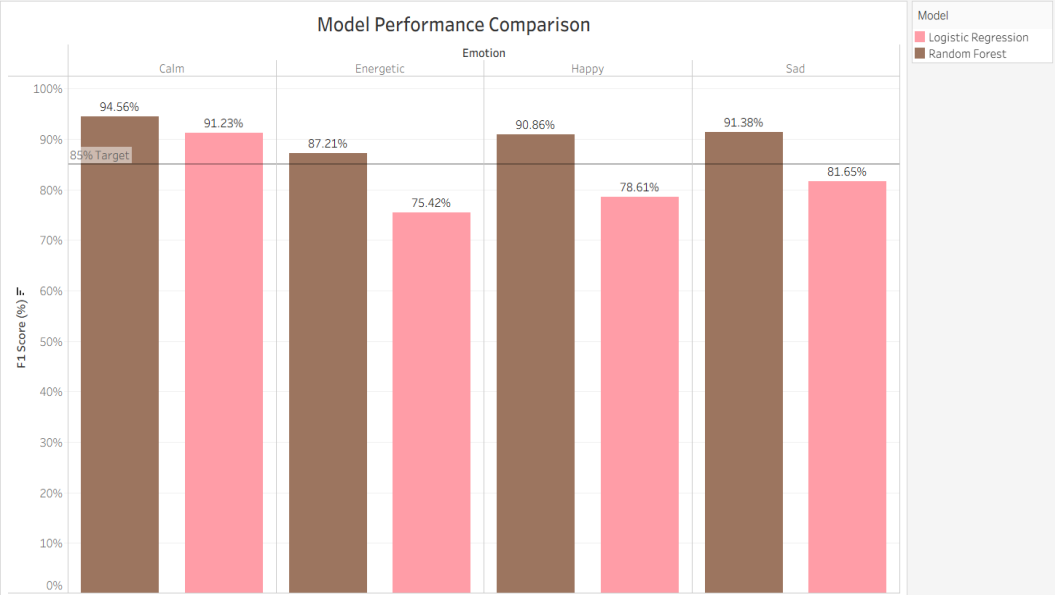
Works for Every Style

It does not get confused by “weird” music.

91% Accuracy

The crowd is almost always right.

The Result



Proven Reliable

Beating the 85% Target
Our model stays above professional standard for every single emotion

The Winning Edge

Outperforming the Standard
Using 100 “expert voters”, our model is significantly more accurate than standard industry tools

The Question

Can a free, open-source Machine Learning replicate commercial emotion tagging at **>= 85% accuracy**?

The Answer

YES! 91% Accuracy

Tag 1,000 songs in under 2 hours
$0 cost. No subscriptions

Your music is finally searchable!

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Emotions Decoded

Sad

acousticness (0.58),
liveness (0.18),
tempo (114.71 BPM)

Happy

danceability (0.68),
valence (0.60),
energy (0.69)

Energetic

energy (0.87),
tempo (133.96 BPM),
liveness (0.24)

Calm

instrumentalness (0.88),
acousticness (0.84),
speechiness (0.06)

Step 4: The Proof

Real-World Validation

Emotions Decoded

Sad

acousticness (0.58),
liveness (0.18),
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Step 5: Your Sync Roadmap

Your catalog is no longer invisible

Create

Make music your way

Tag

Run it through the model in under 2 hours, **free!**

Submit

Upload a searchable, mood-tagged catalog to sync platforms

One time setup, takes minutes

Create

- Make music your way
- Upload to Spotify via DistroKid or TuneCore
- Audio features are automatically captured by Spotify
- Export your features into a spreadsheet

Model tells you why each tag was given

Tag

- **Run it through the model in under 2 hours, free!**
- Spot-check Happy and Energetic, these two overlap most
- Add human review for high-stakes submissions
- *Full transparency*

From invisible to searchable

Submit

- Upload your tagged catalog to MusicBed, Artlist or Pond5 (Sync platforms)
- Music supervisors search by mood
- Sign the agreement, collect your sync fee and royalties
- Monitor which emotion tags lead to placements

Honest Limitations

AI-Generated Labels

Not human listeners, spot check still matters

The “Twins” Problem

Happy and Energetic songs are similar

Spotify API Changes

Restricted for new users. Pivot to Librosa, free library for audio analysis

4 Emotion Scope

Currently four. Can retrain for any custom emotion set

When to use human judgement?

Other Limitations

1 in 11 songs

Model is not perfect

For high-stakes placements like TV commercials or film, spot-check the model's work.

Sound ≠ Story

Model hears the music

It does not read the lyrics. If the words matter to the scene, human review is essential.

From Studio to Screen

One Placement. One tool. Zero Cost.

The barrier is gone

Your music is now searchable